

Section: Particles and Waves

Topic: Interference

Light from a laser passes through two narrow slits and falls on a screen.
An interference pattern of bright and dark fringes is produced.

(a) State the name of this type of interference pattern.

✓ This is called a **Young's fringes** or a **double-slit interference pattern**.

(b) State what is meant by coherent light sources.

✓ **Coherent sources** are sources that:

Emit waves of the **same frequency** and a **constant phase difference**.

This coherence ensures a **stable interference pattern**.

(c) Explain why bright and dark fringes are observed on the screen.

✓ Bright fringes occur where **constructive interference** takes place — waves arrive **in phase**.

Dark fringes occur where **destructive interference** happens — waves arrive **out of phase** by π radians (or half a wavelength).

The alternating bright and dark fringes result from the **path difference** between the waves from the two slits.

🧠 Revision Tips

- **Coherence** is essential for a stable interference pattern
- Bright fringe: **constructive interference** → path difference = $n\lambda$
- Dark fringe: **destructive interference** → path difference = $(n + \frac{1}{2})\lambda$
- The pattern depends on **wavelength**, **slit separation**, and **distance to the screen**